Upper percentage points of the studenized range (1%)

	p = number of treatment										
df	2	3	4	5	6	7	8	9	10	11	12
1	90.020	135.000	164.300	185.600	202.200	215.800	227.200	237.000	245.600	253.200	260.000
2	14.040	19.020	22.290	24.720	26.630	28.200	29.530	30.680	31.690	32.590	33.400
3	8.260	10.620	12.170	13.320	14.240	15.000	15.650	16.210	16.690	17.130	17.530
4	6.511	8.120	9.173	9.958	10.580	11.100	11.540	11.920	12.260	12.570	12.840
5	5.702	6.976	7.804	8.421	8.913	9.321	9.669	9.971	10.240	10.480	10.700
6	5.243	6.331	7.033	7.556	7.972	8.318	8.612	8.869	9.097	9.300	9.485
7	4.949	5.919	6.542	7.005	7.373	7.678	7.939	8.166	8.367	8.548	8.711
8	4.745	5.635	6.204	6.625	6.959	7.237	7.474	7.680	7.863	8.027	8.176
9	4.596	5.428	5.957	6.347	6.657	6.915	7.134	7.325	7.494	7.646	7.784
10	4.482	5.270	5.769	6.136	6.428	6.669	6.875	7.054	7.213	7.356	7.485
11	4.392	5.146	5.621	5.970	6.247	6.476	6.671	6.841	6.992	7.127	7.250
12	4.320	5.046	5.502	5.836	6.101	6.320	6.507	6.670	6.814	6.943	7.060
13	4.260	4.964	5.404	5.726	5.981	6.192	6.372	6.528	6.666	6.791	6.903
14	4.210	4.895	5.322	5.634	5.881	6.085	6.258	6.409	6.543	6.663	6.772
15	4.167	4.836	5.252	5.556	5.796	5.994	6.162	6.309	6.438	6.555	6.660
16	4.131	4.786	5.192	5.489	5.722	5.915	6.079	6.222	6.348	6.461	6.564
17	4.099	4.742	5.140	5.430	5.659	5.847	6.007	6.147	6.270	6.380	6.480
18	4.071	4.703	5.094	5.379	5.603	5.787	5.944	6.081	6.201	6.309	6.407
19	4.046	4.669	5.054	5.334	5.553	5.735	5.889	6.022	6.141	6.246	6.342
20	4.024	4.639	5.018	5.293	5.510	5.688	5.839	5.970	6.086	6.190	6.285
24	3.955	4.546	4.907	5.168	5.373	5.542	5.685	5.809	5.919	6.017	6.105
30	3.889	4.455	4.799	5.048	5.242	5.401	5.536	5.653	5.756	5.848	5.932
40	3.825	4.367	4.695	4.931	5.114	5.265	5.392	5.502	5.599	5.685	5.764
60	3.762	4.282	4.594	4.818	4.991	5.133	5.253	5.356	5.447	5.528	5.601
120	3.702	4.200	4.497	4.708	4.872	5.005	5.118	5.214	5.299	5.375	5.443
1000	3.643	4.120	4.403	4.603	4.757	4.882	4.987	5.078	5.157	5.227	5.290

Upper percentage points of the studenized range (1%)

	p = number of treatment												
df	13	14	15	16	17	18	19	20	30	40	60	80	100
1	266.200	271.800	277.000	281.800	286.300	290.400	294.300	298.000	326.000	344.800	370.100	387.300	400.100
2	34.130	34.810	35.430	36.000	36.530	37.030	37.500	37.950	41.320	43.610	46.700	48.800	50.380
3	17.890	18.220	18.520	18.810	19.070	19.320	19.550	19.770	21.440	22.590	24.130	25.190	25.990
4	13.090	13.320	13.530	13.720	13.910	14.080	14.240	14.390	15.570	16.380	17.460	18.200	18.770
5	10.890	11.080	11.240	11.400	11.550	11.680	11.810	11.930	12.870	13.510	14.390	14.990	15.450
6	9.653	9.808	9.951	10.080	10.210	10.320	10.430	10.540	11.340	11.890	12.650	13.170	13.550
7	8.860	8.997	9.124	9.242	9.353	9.456	9.553	9.645	10.360	10.850	11.520	11.980	12.340
8	8.311	8.436	8.552	8.659	8.760	8.854	8.943	9.027	9.677	10.130	10.740	11.170	11.490
9	7.910	8.025	8.132	8.232	8.325	8.412	8.495	8.573	9.177	9.594	10.170	10.560	10.860
10	7.603	7.712	7.812	7.906	7.993	8.075	8.153	8.226	8.794	9.186	9.726	10.100	10.380
11	7.362	7.464	7.560	7.648	7.731	7.809	7.883	7.952	8.491	8.864	9.377	9.732	10.000
12	7.166	7.265	7.356	7.441	7.520	7.594	7.664	7.730	8.246	8.602	9.093	9.433	9.693
13	7.006	7.100	7.188	7.269	7.345	7.417	7.484	7.548	8.043	8.386	8.859	9.186	9.436
14	6.871	6.962	7.047	7.125	7.199	7.268	7.333	7.394	7.873	8.204	8.661	8.978	9.219
15	6.756	6.845	6.927	7.003	7.074	7.141	7.204	7.264	7.727	8.049	8.492	8.800	9.034
16	6.658	6.744	6.823	6.897	6.967	7.032	7.093	7.151	7.602	7.915	8.346	8.646	8.874
17	6.572	6.656	6.733	6.806	6.873	6.937	6.997	7.053	7.493	7.798	8.219	8.511	8.734
18	6.496	6.579	6.655	6.725	6.791	6.854	6.912	6.967	7.397	7.696	8.107	8.393	8.611
19	6.430	6.510	6.585	6.654	6.719	6.780	6.837	6.891	7.312	7.605	8.008	8.288	8.501
20	6.370	6.449	6.523	6.591	6.654	6.714	6.770	6.823	7.237	7.523	7.919	8.194	8.404
24	6.186	6.261	6.330	6.394	6.453	6.510	6.562	6.612	7.001	7.270	7.641	7.900	8.097
30	6.008	6.078	6.142	6.202	6.258	6.311	6.361	6.407	6.771	7.023	7.370	7.611	7.796
40	5.835	5.900	5.961	6.017	6.069	6.118	6.165	6.208	6.547	6.781	7.104	7.328	7.499
60	5.667	5.728	5.784	5.837	5.886	5.931	5.974	6.015	6.329	6.546	6.843	7.049	7.207
120	5.505	5.561	5.614	5.662	5.708	5.750	5.790	5.827	6.117	6.316	6.588	6.776	6.919
1000	5.348	5.400	5.448	5.493	5.535	5.574	5.611	5.645	5.911	6.092	6.338	6.507	6.636

Upper percentage points of the studenized range (5%)

	p = number of treatment										
df	2	3	4	5	6	7	8	9	10	11	12
1	17.970	26.980	32.820	37.080	40.410	43.120	45.400	47.360	49.070	50.590	51.960
2	6.085	8.331	9.799	10.880	11.730	12.430	13.030	13.540	13.990	14.400	14.760
3	4.501	5.910	6.825	7.502	8.037	8.478	8.852	9.177	9.462	9.717	9.946
4	3.926	5.040	5.757	6.287	6.706	7.053	7.347	7.602	7.826	8.027	8.208
5	3.635	4.602	5.218	5.673	6.033	6.330	6.582	6.801	6.995	7.167	7.323
6	3.460	4.339	4.896	5.305	5.628	5.895	6.122	6.319	6.493	6.649	6.789
7	3.344	4.165	4.681	5.060	5.359	5.606	5.815	5.997	6.158	6.302	6.431
8	3.261	4.041	4.529	4.886	5.167	5.399	5.596	5.767	5.918	6.053	6.175
9	3.199	3.948	4.415	4.755	5.024	5.244	5.432	5.595	5.738	5.867	5.983
10	3.151	3.877	4.327	4.654	4.912	5.124	5.304	5.460	5.598	5.722	5.833
11	3.113	3.820	4.256	4.574	4.823	5.028	5.202	5.353	5.486	5.605	5.713
12	3.081	3.773	4.199	4.508	4.750	4.950	5.119	5.265	5.395	5.510	5.615
13	3.055	3.734	4.151	4.453	4.690	4.884	5.049	5.192	5.318	5.431	5.533
14	3.033	3.701	4.111	4.407	4.639	4.829	4.990	5.130	5.253	5.364	5.463
15	3.014	3.673	4.076	4.367	4.595	4.782	4.940	5.077	5.198	5.306	5.403
16	2.998	3.649	4.046	4.333	4.557	4.741	4.896	5.031	5.150	5.256	5.352
17	2.984	3.628	4.020	4.303	4.524	4.705	4.858	4.991	5.108	5.212	5.306
18	2.971	3.609	3.997	4.276	4.494	4.673	4.824	4.955	5.071	5.173	5.266
19	2.960	3.593	3.977	4.253	4.468	4.645	4.794	4.924	5.037	5.139	5.231
20	2.950	3.578	3.958	4.232	4.445	4.620	4.768	4.895	5.008	5.108	5.199
24	2.919	3.532	3.901	4.166	4.373	4.541	4.684	4.807	4.915	5.012	5.099
30	2.888	3.486	3.845	4.102	4.301	4.464	4.601	4.720	4.824	4.917	5.001
40	2.858	3.442	3.791	4.039	4.232	4.388	4.521	4.634	4.735	4.824	4.904
60	2.829	3.399	3.737	3.977	4.163	4.314	4.441	4.550	4.646	4.732	4.808
120	2.800	3.356	3.685	3.917	4.096	4.241	4.363	4.468	4.560	4.641	4.714
1000	2.772	3.314	3.633	3.858	4.030	4.170	4.286	4.387	4.474	4.552	4.622

Upper percentage points of the studenized range (5%)

	p = number of treatment												
df	13	14	15	16	17	18	19	20	30	40	60	80	100
1	53.200	54.330	55.360	56.320	57.220	58.040	58.830	59.560	65.150	68.920	73.970	77.400	79.980
2	15.090	15.390	15.650	15.920	16.140	16.380	16.570	16.780	18.270	19.280	20.660	21.590	22.290
3	10.150	10.350	10.520	10.690	10.840	10.980	11.110	11.240	12.210	12.860	13.760	14.360	14.820
4	8.373	8.524	8.664	8.793	8.914	9.027	9.133	9.233	10.000	10.530	11.240	11.730	12.100
5	7.466	7.596	7.716	7.828	7.932	8.030	8.122	8.208	8.875	9.330	9.949	10.370	10.690
6	6.917	7.034	7.143	7.244	7.338	7.426	7.508	7.586	8.189	8.601	9.162	9.547	9.839
7	6.550	6.658	6.759	6.852	6.939	7.020	7.097	7.169	7.727	8.110	8.631	8.989	9.260
8	6.287	6.389	6.483	6.571	6.653	6.729	6.801	6.869	7.395	7.756	8.248	8.586	8.843
9	6.089	6.186	6.276	6.359	6.437	6.510	6.579	6.643	7.144	7.488	7.958	8.281	8.526
10	5.935	6.028	6.114	6.194	6.269	6.339	6.405	6.467	6.948	7.278	7.730	8.041	8.276
11	5.811	5.901	5.984	6.062	6.134	6.202	6.265	6.325	6.790	7.109	7.546	7.847	8.075
12	5.710	5.797	5.878	5.953	6.023	6.089	6.151	6.209	6.660	6.970	7.394	7.687	7.908
13	5.625	5.711	5.789	5.862	5.931	5.995	6.055	6.112	6.551	6.853	7.267	7.552	7.769
14	5.554	5.637	5.714	5.785	5.852	5.915	5.973	6.029	6.459	6.754	7.159	7.437	7.649
15	5.492	5.574	5.649	5.719	5.785	5.846	5.904	5.958	6.379	6.669	7.065	7.338	7.546
16	5.439	5.519	5.593	5.662	5.726	5.786	5.843	5.896	6.310	6.594	6.983	7.252	7.456
17	5.392	5.471	5.544	5.612	5.675	5.734	5.790	5.842	6.249	6.529	6.912	7.176	7.377
18	5.351	5.429	5.501	5.567	5.629	5.688	5.743	5.794	6.195	6.471	6.848	7.108	7.307
19	5.314	5.391	5.462	5.528	5.589	5.647	5.701	5.752	6.147	6.419	6.791	7.048	7.244
20	5.282	5.357	5.427	5.492	5.553	5.610	5.663	5.714	6.104	6.372	6.740	6.994	7.187
24	5.179	5.251	5.319	5.381	5.439	5.494	5.545	5.594	5.968	6.226	6.578	6.822	7.007
30	5.077	5.147	5.211	5.271	5.327	5.379	5.429	5.475	5.833	6.080	6.417	6.650	6.827
40	4.977	5.044	5.106	5.163	5.216	5.266	5.313	5.358	5.700	5.934	6.255	6.477	6.645
60	4.878	4.942	5.001	5.056	5.107	5.154	5.199	5.241	5.566	5.789	6.093	6.302	6.462
120	4.781	4.842	4.898	4.950	4.998	5.043	5.086	5.126	5.434	5.644	5.929	6.126	6.275
1000	4.685	4.743	4.796	4.845	4.891	4.934	4.974	5.012	5.301	5.498	5.764	5.947	6.085